

Auto-Encoders for Unsupervised Industrial Anomaly Detection

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Abstract—This report evaluates unsupervised generative models for automated visual inspection and defect localization. We compare Convolutional Autoencoders (CAE) and Adversarial Autoencoders (AAE) trained on nominal distributions to reconstruct defect-free structures. Unseen anomalies are isolated using pixel-precise reconstruction errors. We describe our model architectures, optimization adjustments, and threshold routines aligned with industrial benchmarks.

Index Terms—Anomaly detection, Convolutional Autoencoders, Adversarial Autoencoders, MVTec AD, structural defect localization.

I. INTRODUCTION

Automated visual quality control is vital for modern manufacturing lines to improve efficiency, maintain rigorous production standards, and reduce manual labor overhead. However, detecting defects on complex surfaces poses a significant computer vision challenge due to data asymmetry: anomaly-free samples are abundant, while defects are rare, unpredictable, and highly diverse. Supervised approaches, which require balanced classes and exhaustive labeling of every potential fault, are fundamentally impractical for industrial deployment since unexpected defects can appear at any stage of production.

To solve this, we implement an unsupervised generative framework. By training exclusively on nominal data, the networks learn the standard distribution of textures and object geometries. During testing, the models fail to reconstruct unseen anomalous features, resulting in high pixel-wise reconstruction errors that enable accurate defect segmentation. This framework shifts the problem from multi-class defect classification to a general outlier detection task, eliminating the need for anomalous training samples.

II. RELATED WORK

Our work focuses on deep generative models, comparing a Convolutional Autoencoder (CAE) baseline with an Adversarial Autoencoder (AAE) using Mean Squared Error (MSE) and Multi-Scale Structural Similarity (MS-SSIM) loss functions. The CAE compresses the input into a compact latent space before reconstruction. The AAE introduces a latent discriminator to regularize the feature distribution against a continuous Gaussian prior, ensuring smooth mappings and

reducing spatial blurring. While feature-embedding methods using pre-trained networks provide competitive localization, generative reconstruction remains highly transparent, mapping reconstruction failures directly to the physical location of surface defects.

III. DATASET DESCRIPTION

Adapting standard classification datasets (e.g., MNIST, CIFAR-10) for anomaly detection evaluates outlier detection at a global image level rather than simulating real industrial inspection, where defects are subtle and localized. The MVTec AD dataset offers diverse texturized surfaces and manufactured objects under controlled industrial conditions, complete with pixel-accurate ground truth masks.

The MVTec AD benchmark contains 15 categories with 3,629 images for training/validation and 1,725 for testing. The training partition contains only defect-free samples, while the test set includes 73 different defect types (e.g., scratches, stains, dents, cracks, missing sub-parts). The dataset covers two domains: textures (regular and random arrangements) and objects (rigid components, deformable cables, and items with varying alignment poses). Three categories consist of single-channel gray-scale images, while the rest are high-resolution RGB images scaled for analysis.

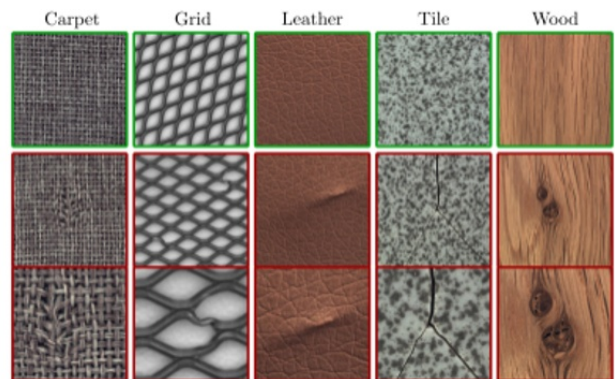


Fig. 1. Examples of regular structural arrangements and natural texture anomalies (Carpet, Grid, Leather, Tile, Wood) from the MVTec AD dataset.

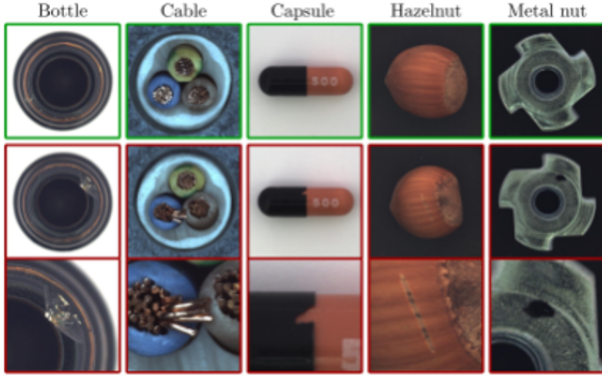


Fig. 2. Examples of manufacturing components exhibiting rigid and deformable structural distortions (Bottle, Cable, Capsule, Hazelnut, Metal nut).

IV. BENCHMARK

A. Evaluated Methods

Our implementation standardizes all image inputs to a target resolution of 256×256 pixels with three-channel RGB data.

1) *Convolutional Autoencoder (CAE) Configuration:* The baseline CAE maps the input tensor through a 7-layer strided convolutional encoder. The channel depth progresses from 3 to 32, 64, 128, 256, 512, and finally 1024, removing the need for explicit max-pooling layers that discard spatial coordinate data. This high-dimensional map is passed through two mirroring dense layers to establish a bottleneck layer. The reconstruction is synthesized via a symmetric transposed convolutional decoder. To amplify backpropagated gradients across subtle, low-contrast boundaries under the L_2 loss configuration, we apply a rigid scaling factor of 10.0 to both the input and reconstructed images directly within the Mean Squared Error objective function.

2) *Adversarial Autoencoder (AAE) Configuration:* The AAE variant routes the encoder output through an intermediate 512-dimensional layer to compute the mean and log-variance vectors for latent sampling via the reparameterization trick. A 1D latent discriminator regularizes this space, acting as a binary classifier to align the latent distribution with the Gaussian prior. This discriminator is structured as a sequence of linear mappings using LeakyReLU and Sigmoid activation functions to output a singular scalar probability. Optimization follows three strict phases: joint encoder-decoder reconstruction updates, discriminator binary cross-entropy minimization to separate real and synthetic vectors, and adversarial encoder training to fool the discriminator.

B. Data Augmentation

To expand the training pool to 10,000 patches per category, we extract random rotated rectangular patches for textures, and apply translations, rotations, and bilateral mirroring for objects based on class geometric symmetries. For complex textures like wood or grid, patches are sampled at multiple scales to make the network robust against localized pattern variations without altering the global structural arrangement.

C. Evaluation Metric

Both architectures evaluate test images by generating a continuous anomaly map, computed as the absolute pixel-wise reconstruction error. To monitor validation performance and calibrate the decision boundaries without data leakage, the nominal dataset is partitioned into an 80% training set and a 20% validation set.

Our evaluation pipeline combines both threshold-independent and threshold-dependent metrics. The threshold-independent quality of the anomaly maps is measured using the pixel-level Area Under the ROC curve (ROC AUC). Additionally, we report the Area Under the Per-Region Overlap (AU-PRO) curve, which weighs small and large defect regions equally to prevent statistical evaluation bias caused by defect size.

For practical industrial application, these continuous maps must be converted into a binary accept/reject decision. To determine the optimal decision boundary, we implement an iterative binary search routine on the validation set maps. For each candidate threshold, a binary mask is generated, and localized anomalous structures are isolated using morphological connected-component labeling.

To filter sensor noise and avoid false rejections, an isolated connected component is flagged as a true anomaly only if its total area exceeds a minimum pixel threshold ($A_{\min}$). This threshold is adapted dynamically to accommodate the different image resolutions of the object categories under analysis. We define a baseline reference where a minimum defect area of 50 pixels is assigned to a standard image size of 900×900 . For images of different dimensions, this reference area is scaled proportionally according to the following formula:

$$A_{\min} = \max \left(1, \left\lfloor \frac{\text{Width} \times \text{Height}}{900^2} \times 50 \right\rfloor \right) \quad (1)$$

A major limitation of this approach lies in the manual selection of this minimum defect area baseline. Because this parameter is a fixed heuristic rather than a data-driven, adaptive statistical threshold, it directly constrains the flexibility of the post-processing pipeline. Setting this value manually introduces a strict trade-off: a baseline that is too low increases false positives from minor illumination variants, while a baseline that is too high completely blinds the system to fine, critical defects like small cracks or scratches.

V. RESULTS AND DISCUSSION

Architectural Performance Trade-offs: Quantitative benchmarks (Table 1 and Table 2) reveal clear trade-offs. The CAE handles large, diffuse anomalies well due to its quadratic loss bias, though it tends to blur sharp edges. The AAE preserves details and isolates fine cracks or scratches effectively because its adversarial constraint enforces a continuous latent space. Removing the post-processing scaling divisor preserves the true error map scale, allowing morphological filters to accurately track gradient boundaries. Object-level accuracy confirms that the choice of architecture depends heavily on target defect morphology.

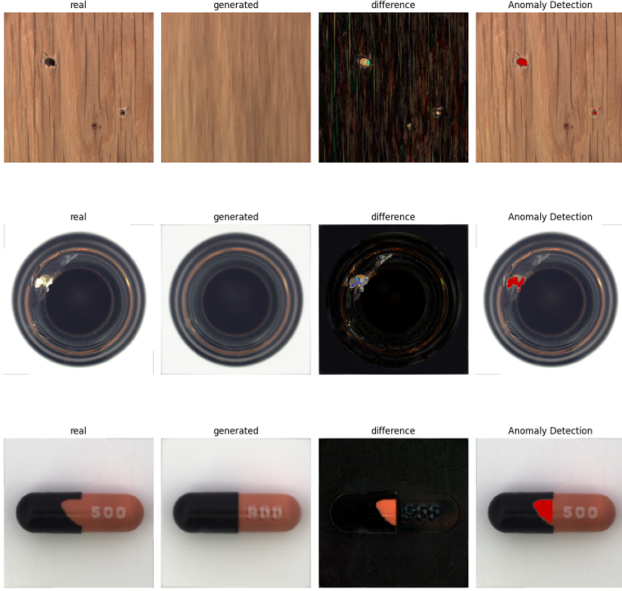


Fig. 3. Examples of obtained results for wood, bottle, capsule.

Analysis of Category Failures: The low classification scores observed in the top rows of Table 1 for specific configurations (e.g., CAE L2 on Bottle yielding 0.00) stem from a structural limitation in autoencoder binarization rather than poor representation learning. Because the model minimizes pixel-wise differences globally, it struggles to completely remove sharp reflections or natural variations on highly reflective materials like glass. This creates minor reconstruction residuals in normal samples. Since the automated threshold routine optimizes globally to maximize anomalous detection, these nominal variations cross the decision boundary, causing the model to misclassify all nominal samples as anomalous (100% false positive rate), while perfectly capturing true defects (1.00 accuracy on anomalous images).

Conversely, for complex structural defects (e.g., the Capsule category), structural shifts or internal alignment variations cause global reconstruction blur. When evaluating with MS-SSIM loss functions, the network prioritizes structural coherence over individual pixel alignment. This results in smoother anomaly maps that lower pixel-level ROC AUC scores but yield much cleaner and more stable segmentation regions (higher AU-PRO) on small defects like scratches or localized stains.

Optimization and Gradient Stabilization: Joint variational-adversarial training requires strict constraints to prevent latent space collapse. To stop extreme gradient updates from violating cross-entropy boundaries, we implement a mathematical bounding function ($\max(\epsilon, \min(1 - \epsilon, D(z)))$ with $\epsilon = 10^{-7}$), a 1.0 gradient clipping norm, bidirectional label smoothing (0.9/0.1), and an asymmetric 1:5 learning rate split ($\eta_{\text{disc}} = \eta_{\text{base}}/5$). Scaling the encoder’s generative loss update to 0.05 ensures smooth feature learning. We

combine instance normalization in the encoder to remove style components with batch normalization in the decoder to stabilize pixel synthesis.

Post-Processing and Structural Similarity: Accurate binarization relies heavily on the loss formulation and adaptive filtering. We integrate a custom Multi-Scale Structural Similarity (MS-SSIM) loss evaluating 5 pooling levels with normalized Wang weights. To prevent numerical instability from fractional exponents on negative values under extreme saturation, we include positive range corrections and a ReLU operator before power functions. This keeps the anomaly map immune to global lighting shifts while delivering precise real-time localization.

Table I

Object	MVTec Baseline			Our Method			
	CAE (SSIM)	CAE (L2)	Ano-GAN	CAE (L2)	CAE (MS.)	AAE (L2)	AAE (MS.)
Bottle	0.85	0.71	0.95	0.00	0.45	0.00	0.55
	0.90	0.89	0.43	1.00	0.94	1.00	0.91
Capsule	0.78	1.00	0.96	0.61	0.22	0.04	0.52
	0.43	0.24	0.20	0.62	0.85	0.98	0.56
Wood	0.84	1.00	0.89	1.00	0.26	0.26	0.21
	0.82	0.47	0.47	0.73	0.93	1.00	0.95

Table I: For each dataset category, the ratio of correctly classified samples of anomaly-free (top row) and anomalous images (bottom row) is given. The method with the highest mean of these two values is highlighted in boldface for each texture and object category.

Table II

Object	MVTec Baseline			Our Method			
	CAE (SSIM)	CAE (L2)	Ano-GAN	CAE (L2)	CAE (MS.)	AAE (L2)	AAE (MS.)
Bottle	0.15	0.22	0.05	0.59	0.57	0.56	0.60
	0.93	0.86	0.86	0.89	0.91	0.88	0.84
Capsule	0.09	0.11	0.04	0.47	0.48	0.11	0.35
	0.94	0.88	0.84	0.66	0.53	0.59	0.53
Wood	0.36	0.29	0.14	0.39	0.38	0.45	0.39
	0.73	0.73	0.62	0.95	0.70	0.88	0.56

Table II: For each dataset category, the AU-PRO (top row) and the ROC AUC (bottom row) are given. The best performing method is highlighted in boldface.

VI. CONCLUSIONS

This work demonstrates that both Convolutional Autoencoders (CAE) and Adversarial Autoencoders (AAE) provide a viable, unsupervised framework for industrial anomaly detection on the MVTec AD dataset. By learning the nominal distribution of defect-free samples, both architectures successfully isolate anomalous structures purely through pixel-wise reconstruction residuals.

However, our analysis highlights that the primary operational bottleneck of the system does not reside within the deep learning architectures themselves, but rather in the deterministic post-processing stage. The final accept/reject decision depends entirely on a static thresholding routine and

a rigid minimum area filter ($A_{\min}$). While this connected-component filtering is necessary to suppress sensor noise and minor camera grain, its geometric assumptions introduce severe limitations. Because industrial defects vary widely in morphology—ranging from fine, low-pixel cracks to large, diffuse stains—a fixed spatial heuristic fails to generalize across different material categories. Consequently, the system faces a strict trade-off: a threshold conservative enough to eliminate background noise will inevitably discard true, subtle structural defects.

Therefore, the final obtainable classification accuracy of this pipeline is inherently bounded by these post-processing heuristics rather than the feature-extraction capacity of the networks. To overcome these constraints, future work will focus on replacing manual geometric rules with adaptive, category-specific density estimation frameworks. By statistically modeling the distribution of the anomaly maps, the system can differentiate true surface defects from sensor noise based on probabilistic density rather than arbitrary pixel counts.

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